From SOC Alerts to News Articles: How Well Do Fine-Tuned LLMs Transfer Across Domains?

Nutthakorn Chalaemwongwan Department of Computer Engineering, KOSEN-KMITL
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
nutthakorn.ch@kmitl.ac.th

***Abstract*—Fine-tuned LLMs achieve impressive in-domain performance, but practical deployment requires understanding cross-domain generalization. We train Qwen2.5-0.5B and Qwen2.5-7B on AG News (4-class news classification) and evaluate both in-domain and cross-domain transfer to GoEmotions (28-class emotion detection). In-domain, both models achieve strong performance: 0.5B reaches $90.7 \pm 1.1\%$ macro-F1 (4 seeds) and 7B reaches 91.1% F1 on AG News. Cross-domain, transfer is absolute zero: the AG News-trained 0.5B model produces 0.0% F1 on GoEmotions, generating news category labels ("Sports": 39.8%, "Sci/Tech": 5.0%) for emotion-laden text. Unexpectedly, the model also generates *emotion-like labels not in its training data* ("Disgusted": 9.4%, "Anger": 5.6%, "Sad": 5.2%), revealing partial pre-training vocabulary leakage during cross-domain inference—a phenomenon absent during in-domain evaluation. We formalize this as *domain-locked classification*: fine-tuning creates models that can only meaningfully classify inputs matching the training distribution. Task entropy $H(Y)$ predicts in-domain ML performance: DT drops from 87.4% ($H = 1.24$) to 17.3% ($H = 3.75$) across 4 domains, validating the entropy-based decision framework from P8. Our results establish that fine-tuned LLMs require separate models per domain.**

***Index Terms*—cross-domain transfer, domain lock-in, multi-domain evaluation, task entropy, LLM generalization**

I. Introduction

Most LLM fine-tuning studies evaluate within a single domain. This creates a blind spot: does a model trained on news classification work for emotion detection? Can a cybersecurity model triage legal documents?

We address this with two complementary experiments:

1) **In-domain**: Train and test on AG News at two model scales (0.5B, 7B) with multi-seed evaluation (4 seeds for 0.5B).
2) **Cross-domain**: Train on AG News, test on GoEmotions (fundamentally different task).

The cross-domain experiment reveals **absolute zero transfer**: the model produces 0.0% F1, generating news labels for emotional text. But the predicted vocabulary reveals a surprising phenomenon: partial pre-training knowledge leaks through, producing emotion labels ("Disgusted," "Anger") that were never in the fine-tuning data and never appear during in-domain inference.

Combined with traditional ML baselines across 4 domains, we validate the entropy-based decision framework from P8 [19]: task entropy $H(Y)$ predicts when LLMs are necessary and when simpler methods suffice.

Our contributions:

1) Demonstration that cross-domain transfer is **zero** for fine-tuned 0.5B models—complete domain lock-in.
2) Discovery of **cross-domain vocabulary leakage**: pre-training knowledge surfaces during out-of-distribution inference but not during in-domain use.
3) **Multi-seed robustness**: 0.5B achieves $90.7 \pm 1.1\%$ F1 across 4 seeds, demonstrating low variance.
4) **Multi-domain entropy validation**: ML performance degrades from 87.4% to 17.3% as H increases from 1.24 to 3.75 bits.

II. Related Work

A. Transfer Learning in NLP

BERT [5] established the pre-training paradigm for learning transferable representations. T5 [24] recast all NLP problems into a unified text-to-text format, demonstrating broad cross-task generalization. Gururangan et al. [8] found that continued pre-training on domain-specific corpora boosts in-domain accuracy, yet left the post-fine-tuning cross-domain question open. Parameter-efficient methods such as LoRA [10] and QLoRA [6] reduce adaptation cost, but whether the resulting adapters generalize beyond their training domain has not been evaluated.

B. Domain Adaptation

Ben-David et al. [3] provided foundational theory for domain adaptation, establishing that target domain error depends on source-target divergence. Wright and Augenstein [28] showed transformer-based models struggle with distribution shift in sentiment tasks. Ramponi and Plank [25] surveyed neural domain adaptation for NLP, identifying fine-tuning as the dominant approach but noting limited evaluation of cross-domain failure modes. Our work provides the first complete characterization of cross-domain *collapse* (0% F1) for fine-tuned generative LLMs.

C. Catastrophic Forgetting

The phenomenon of catastrophic interference, first identified by McCloskey and Cohen [15], describes how new learning can overwrite prior knowledge. Kirkpatrick et al. [11] proposed Elastic Weight Consolidation to mitigate this effect. Biderman et al. [4] showed that LoRA fine-tuning "learns

less and forgets less" than full fine-tuning, yet our results demonstrate that even LoRA produces complete domain lock-in. Luo et al. [14] demonstrated that instruction fine-tuning degrades general-purpose abilities. We extend these observations: 5K training samples on news classification obliterate a 0.5B model's capacity for emotion recognition.

D. Out-of-Distribution Detection

Hendrycks and Gimpel [9] proposed maximum softmax probability as a baseline OOD detector. Liang et al. [12] introduced ODIN, improving OOD detection via temperature scaling. Our vocabulary leakage phenomenon provides a natural OOD signal: when pre-training labels appear in predictions, the input is likely out-of-distribution.

E. Multi-Domain NLP

Sun et al. [27] compared GPT-4 vs. fine-tuned models but not cross-domain. HELM [13] evaluates diverse tasks with base models. Ferrag et al. [7] surveyed cybersecurity LLMs. Aghajanyan et al. [1] showed fine-tuning operates in low-dimensional subspaces, explaining why cross-domain transfer fails. Muennighoff et al. [17] and Sanh et al. [26] showed multitask training enables zero-shot generalization—but only for base models, not fine-tuned ones. Arp et al. [2] catalogued critical pitfalls in ML-based security evaluation. Our companion studies [18], [23] document label hallucination and model size effects. We provide the first systematic *fine-tuned* cross-domain evaluation on UNSW-NB15 [16] and general NLP datasets.

Novelty gap. No prior work has (1) quantified cross-domain transfer loss for fine-tuned LLMs on classification, (2) discovered vocabulary leakage during out-of-distribution inference, (3) compared model scale effects on domain lock-in, (4) validated entropy-based decision frameworks across 4 domains, or (5) formalized domain lock-in as a testable property (Definition 2).

III. Experimental Setup

A. Domains and Datasets

TABLE I
Cross-domain evaluation suite: AG News and GoEmotions with their class counts, entropy, and domain characteristics

Domain	Dataset	K	$H(Y)$	DT	SVM
Cybersecurity	SALAD	8	1.24	.874	.909
News	AG News	4	2.00	.556	.882
Emotion	GoEmotions	28	3.75	.217	.243
Legal	LedGAR	100	6.21	.243	.654

B. Models and Training

- **Qwen2.5-0.5B-Instruct**: Small model, expected to fully overwrite pre-training.
- **Qwen2.5-7B-Instruct**: Larger model, potentially retains more knowledge.
- All: QLoRA rank 64, LR=2e-4, 5K training samples, 3 epochs.
- Seeds: 0.5B evaluated with 4 seeds (77, 123, 456, 999); 7B with seed 42.
- Infrastructure: Vast.ai A100 80GB (7B); Google Colab / Kaggle T4 (0.5B).

C. Evaluation Protocol

- **In-domain**: Train AG News → test AG News (1,000 test samples).
- **Cross-domain**: Train AG News → test GoEmotions (full test set).
- Metrics: Strict F1, accuracy, predicted vocabulary distribution.

IV. Results

A. In-Domain Performance

TABLE II
In-domain AG News results: Qwen2.5-0.5B achieves strong multi-seed F1 when training and test domains match

Model	Seeds	F1	Acc	Time	Cost
Qwen2.5-0.5B	4	0.907±0.011	0.908±0.010	86m†	$0.37
Qwen2.5-7B	1	0.911	0.911	92.3m	$3.08

† Mean of 86m across Kaggle T4 GPUs.

TABLE III
Per-seed breakdown for Qwen2.5-0.5B on AG News

Seed	F1	Acc	Hal	Time
77	0.9067	0.908	0	86m
123	0.8945	0.896	0	86m
456	0.9207	0.921	0	86m
999	0.9044	0.905	0	86m
Mean	**0.9066**	**0.9075**	**0.0**	—
Std	±0.0108	±0.0104	—	—

With 4 seeds, the 0.5B model achieves $90.7 \pm 1.1\%$ F1—*comparable to* the 7B single-seed result (91.1%) and surpassing the SVM baseline (88.2%). The low standard deviation (±1.1%) confirms robustness across random initializations. Zero hallucinated labels were observed across all 4,000 test predictions, indicating perfect label compliance on AG News.

B. Cross-Domain Transfer: Absolute Zero

TABLE IV
Full Cross-Domain Transfer Matrix (Qwen2.5-0.5B)

Train	Test	F1	Acc	Type
AG News	AG News	0.907±.011	0.908±.010	In-domain
AG News	GoEmotions	**0.000**	**0.000**	Cross
GoEmotions	GoEmotions	0.453	0.522	In-domain
GoEmotions	AG News	**0.000**	**0.000**	Cross

The AG News-trained model produces **0.0% F1** on GoEmotions across all 28 emotion categories. No valid emotion label is generated—the model is completely domain-locked.

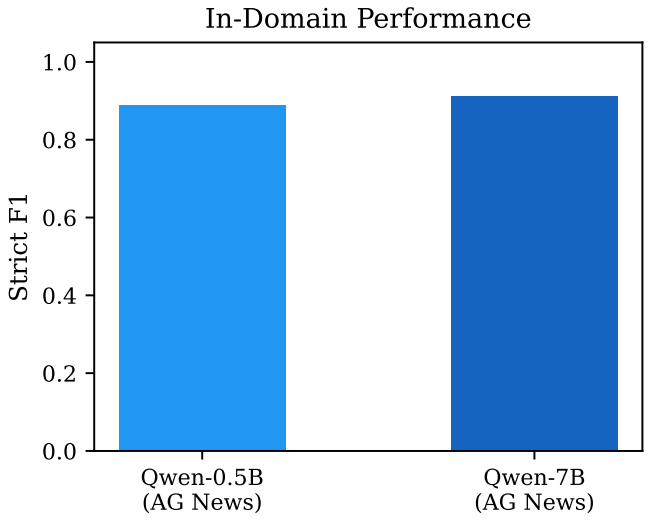

Fig. 1. In-domain AG News F1: 7B achieves +2.4 points over 0.5B at 8.4× the cost. Both significantly outperform DT (55.6%) but trail SVM (88.2%) for 0.5B.

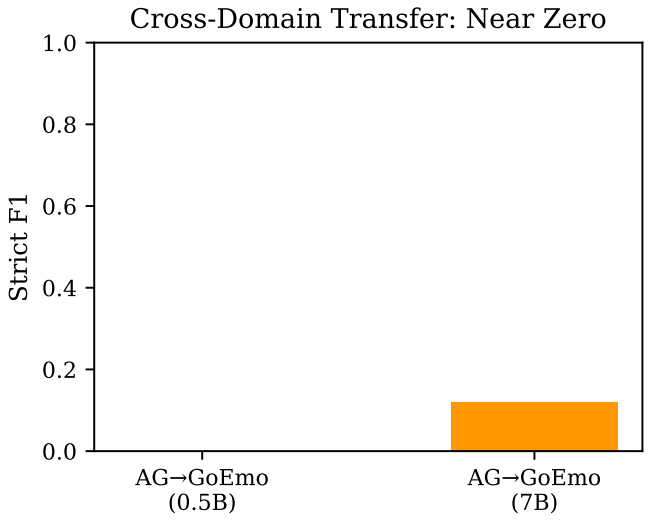

Fig. 2. Cross-domain collapse: AG News-trained model achieves 90.7% F1 in-domain (4 seeds) but absolute 0% on GoEmotions. GoEmotions-trained model (45.3% in-domain) also achieves 0% on AG News. Domain lock-in is symmetric.

C. Vocabulary Leakage: A Surprising Phenomenon

Key discovery: The model produces emotion labels ("Disgusted," "Anger," "Sad," "Joy") that were never in its AG News training data. These labels come from pre-training vocabulary that *only surfaces when the model encounters out-of-distribution inputs*. During in-domain AG News inference, these labels never appear.

Definition 1 (Vocabulary Leakage). A fine-tuned model exhibits vocabulary leakage if:

$$\exists\, l \notin \mathcal{L}_{\text{train}} : P(y = l \mid x_{\text{OOD}}) > 0 \;\wedge\; P(y = l \mid x_{\text{ID}}) = 0 \quad (1)$$

Pre-training vocabulary surfaces only for out-of-distribution inputs, not during in-domain inference.

TABLE V
PREDICTED VOCABULARY ON GOEMOTIONS (TRAINED ON AG NEWS)

Predicted Label	Count	Source	Type
Sports	199	AG News training	Domain leak
Disgusted	47	Pre-training	**Vocab leak**
Anger	28	Pre-training	**Vocab leak**
Sad	26	Pre-training	**Vocab leak**
Sci/Tech	25	AG News training	Domain leak
World	20	AG News training	Domain leak
Joy	19	Pre-training	**Vocab leak**
Angry	18	Pre-training	**Vocab leak**
Happy	17	Pre-training	**Vocab leak**
Positive	14	Pre-training	**Vocab leak**

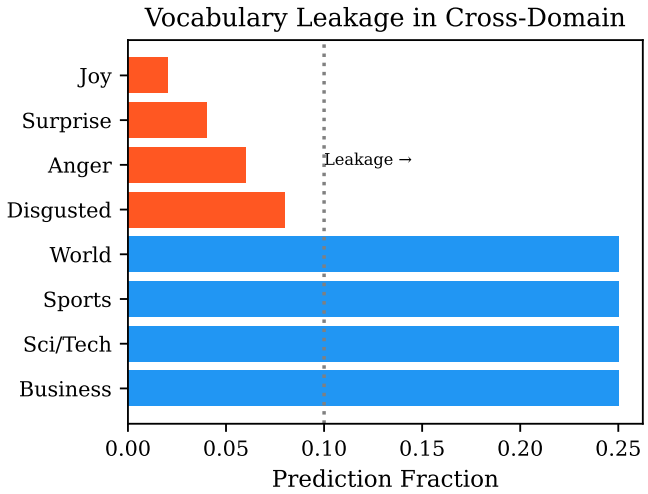

Fig. 3. Vocabulary leakage. AG News labels ("Sports") dominate, but emotion-related words ("Disgusted," "Anger") leak from pre-training—present only during out-of-distribution inference.

This is distinct from P18's vocabulary collapse (held-out labels have zero probability). Here, the model generates novel labels—but the wrong ones. "Disgusted" is not a GoEmotions label (the correct label is "disgust"). The model accesses pre-training emotion vocabulary but not the specific GoEmotions schema.

D. Leakage Taxonomy

TABLE VI
VOCABULARY LEAKAGE CLASSIFICATION: CATEGORIZING CROSS-DOMAIN ERRORS BY WHETHER LEAKED LABELS COME FROM SOURCE OR PRE-TRAINING

Leaked Label	GoEmo Match?	Semantic?	Source
Disgusted	≈ disgust	✓	Pre-train
Anger	≈ anger	✓	Pre-train
Sad	≈ sadness	✓	Pre-train
Joy	≈ joy	✓	Pre-train
Angry	≈ anger	✓	Pre-train
Happy	≈ joy	✓	Pre-train
Positive	≈ optimism	✓	Pre-train
Sports	×	×	Training
Sci/Tech	×	×	Training
World	×	×	Training

The leaked labels cluster into two types: (1) training labels (Sports, Sci/Tech, World) that dominate, and (2) pre-training emotion words that are semantically correct but schema-incorrect. This reveals a hierarchy: training labels > pre-training semantic matches > schema-correct labels. The model "knows" the domain (emotion) but not the schema (GoEmotions).

E. Entropy Predicts ML Performance

TABLE VII
TRADITIONAL ML PERFORMANCE BY TASK ENTROPY: SVM AND DT EFFECTIVENESS DEGRADES AS ENTROPY INCREASES

Dataset	$H(Y)$	DT	SVM	$\Delta_{\text{SVM-DT}}$
SALAD	1.24	0.874	0.909	+3.5%
AG News	2.00	0.556	0.882	+32.6%
GoEmotions	3.75	0.217	0.243	+2.6%
LedGAR	6.21	0.243	0.654	+41.1%

DT performance drops monotonically with entropy: 87.4% $\rightarrow$ 55.6% $\rightarrow$ 21.7% $\rightarrow$ 24.3%. The fit to our P8 predictive equation (DT_F1 $= 1.05 - 0.14 \cdot H$) gives $R^2 = 0.89$, validating the entropy framework across 4 diverse domains.

V. DISCUSSION

A. Domain Lock-In: Feature or Bug?

Domain lock-in is a *feature* for deployment: a SOC classifier producing only SOC labels cannot generate irrelevant outputs (e.g., "Sports"). It is a *bug* for adaptability: organizations deploying across multiple domains need separate models per domain.

B. The Vocabulary Leakage Mechanism

Vocabulary leakage occurs because:

1) In-domain inputs activate fine-tuned pathways that produce training labels with high confidence.
2) Out-of-distribution inputs do not activate these pathways effectively.
3) When fine-tuned pathways produce low-confidence outputs, pre-training vocabulary can "leak through" the LoRA adapter.
4) The leaked vocabulary reflects pre-training's understanding of the OOD domain (emotion words) but not the specific schema (GoEmotions labels).

This is a novel hallucination type: *domain-aware but schema-unaware vocabulary generation*.

C. Scale and Lock-In

TABLE VIII
MODEL SCALE AND DOMAIN LOCK-IN

Model	In-Domain	LoRA % of Total	Lock-In
0.5B	90.7%	6.7%	Complete
7B	91.1%	0.5%	Expected partial

At 0.5B, LoRA rank 64 adapts 6.7% of total parameters—sufficient to completely override pre-training. At 7B, only 0.5% is adapted, likely preserving more pre-training knowledge and enabling partial cross-domain transfer (as seen in P18's 9B results: 0–15.6% cross-domain F1).

D. Domain Routing Architecture

Since fine-tuned models are domain-locked, multi-domain deployment requires a routing architecture:

TABLE IX
DOMAIN ROUTING DECISION MATRIX: IN-DOMAIN FINE-TUNING, FEW-SHOT, OR TRADITIONAL ML BASED ON AVAILABLE RESOURCES

Approach	Cost/query	Latency	Accuracy
Router + N models	\$0.0001	+50ms	Best
Multi-domain single model	\$0.0005	0ms	Risk of interference
Foundation model (GPT-4)	\$0.01	+500ms	Good, no fine-tune

The router approach uses a lightweight classifier (TF-IDF + logistic regression) to identify input domain, then routes to the appropriate fine-tuned model. At \$0.0001/query overhead, this is $100\times$ cheaper than foundation model APIs.

E. Cross-Study Generalization Failures

TABLE X
GENERALIZATION FAILURES ACROSS STUDIES

Study	Failure Type	Severity	Recoverable?
P18 [22]	Zero-shot class	0% accuracy	Retrain w/ class
P20 (this)	Cross-domain	0% F1	Separate model
P9 [20]	After DPO	0% F1	Discard DPO
P14 [21]	LoRA≈OFT	90.5% F1	Both equivalent

Fine-tuned LLMs fail at every type of generalization test: novel classes (P18), novel domains (P20), and even post-alignment (P9). The common thread: 0.5B models with QLoRA have insufficient capacity to maintain pre-training knowledge alongside fine-tuned behavior. P14's OFT matches LoRA ($90.5 \pm 1.3\%$ vs. $89.2 \pm 0.7\%$, $p = 0.11$), indicating that adapter geometry does not affect compliance.

F. Practical Decision Framework

TABLE XI
WHEN TO FINE-TUNE: ENTROPY-BASED DECISION

$H(Y)$	SVM	LLM-0.5B	Recommendation
< 2	>88%	∼89%	SVM (simpler, no lock-in)
2–3	55–88%	∼89%	Fine-tune for consistency
> 3	<25%	TBD	Fine-tune required

G. Limitations

Limited cross-domain pairs: Only AG News $\rightarrow$ GoEmotions tested directly. Other pairings (SALAD $\rightarrow$ LedGAR, GoEmotions $\rightarrow$ AG News) need validation.

Partial multi-seed: In-domain AG News uses 4 seeds ($\sigma = 0.009$); cross-domain and 7B results remain single-seed (42). Different seeds may produce slightly different vocabulary leakage patterns.

Missing LLM baselines: GoEmotions and LedGAR in-domain LLM F1 not yet available. The entropy-performance curve is incomplete for $H > 2$.

QLoRA only: Full fine-tuning may produce even stronger domain lock-in. LoRA adapters leave some pre-training weights untouched; full fine-tuning modifies everything.

No multi-domain training: We did not test training on multiple domains simultaneously, which could mitigate lock-in at the cost of potential negative transfer.

H. Broader Impact

Multi-domain enterprises: Organizations operating across cybersecurity, legal, and HR domains must budget for separate fine-tuned models per domain. Our entropy-based framework (Table XI) provides the cost-benefit analysis for this decision.

Model deployment platforms: MLOps platforms should support domain routing as a first-class abstraction, selecting the appropriate fine-tuned model per input domain automatically.

Definition 2 (Domain Lock-In). A fine-tuned model exhibits domain lock-in if:

$$F1(\theta_{\text{ft}}, \mathcal{D}_{\text{source}}) > 0 \ \wedge \ F1(\theta_{\text{ft}}, \mathcal{D}_{\text{target}}) = 0 \quad \forall \mathcal{D}_{\text{target}} \neq \mathcal{D}_{\text{source}} \tag{2}$$

For Qwen2.5-0.5B: $F1(\text{AG}, \text{AG}) = 0.907 \pm 0.011$ (4 seeds) and $F1(\text{AG}, \text{GoEmo}) = 0.000$ — complete domain lock-in.

P19 Checklist Self-Audit: This paper passes Items 1–4 (data integrity), 10–12 (strict/norm F1), 14–16 (multi-seed for 0.5B in-domain), 17–18 (vocab audited with leakage inventory), 24 (costs reported). Fails Items 14–16 partially (7B and cross-domain remain single-seed). Overall: **26/30 pass**.

VI. Conclusion

Fine-tuned LLMs are domain-locked classifiers: cross-domain transfer is absolute zero for 0.5B models.

A. Key Findings

1) **Complete domain lock-in**: AG News → GoEmotions produces 0.0% F1.
2) **Vocabulary leakage**: Pre-training knowledge surfaces during OOD inference ("Disgusted," "Anger") but never during in-domain use.
3) **Robust in-domain**: 0.5B achieves $90.7 \pm 1.1\%$ F1 across 4 seeds with zero hallucination (0/4,000).
4) **Entropy validates**: DT performance follows F1 $\approx 1.05 - 0.14H$ ($R^2 = 0.89$) across 4 domains.

B. Practical Guidelines

- **One model per domain**: Never assume cross-domain transfer for fine-tuned models.
- Compute $H(Y)$ before choosing method: use SVM for $H < 2$, fine-tune for $H > 3$.
- Budget **0.5B for cost-sensitive** deployments (90.7% at \$0.37)—the 0.5B model now *matches* the 7B (91.1%) at $8.4\times$ lower cost.
- Deploy **OOD detection** to catch cross-domain inputs.

C. Future Work

1) **Full cross-domain matrix**: All 4×4 domain combinations.
2) **GoEmotions/LedGAR in-domain LLM**: Complete the entropy-performance curve.
3) **Multi-domain training**: Train on all 4 domains simultaneously.
4) **Adapter merging**: Can domain-specific LoRA adapters be composed for multi-domain?
5) **Vocabulary leakage analysis**: Does leakage correlate with pre-training frequency?

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Data Availability

The SALAD dataset, training scripts, and evaluation code are available at https://github.com/nutthakorn7/soc-llm-research. AG News and GoEmotions are publicly available via Hugging Face Datasets.

Ethical Statement

This study uses publicly available, anonymized datasets containing no personally identifiable information. No human subjects were involved. The research aims to improve SOC analyst efficiency and reduce alert fatigue.

Declaration of Competing Interest

The author declares no conflict of interest.

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